

Book Inventory Application

Spring Boot REST API Endpoints

Book APIs

HTTP

GET /api/v1/books

GET /api/v1/books/{isbn}

POST /api/v1/books

PUT /api/v1/books/{isbn}

DELETE /api/v1/books/{isbn}

Book Search / Filter

HTTP

GET /api/v1/books?categoryId={catId}&publisherId={publisherId}

Category APIs (Read-Only)

HTTP

GET /api/v1/categories

GET /api/v1/categories/{categoryId}

Author APIs

HTTP

GET /api/v1/authors

GET /api/v1/authors/{authorId}

POST /api/v1/authors

PUT /api/v1/authors/{authorId}

DELETE /api/v1/authors/{authorId}

Book–Author Mapping

HTTP

POST /api/v1/books/{isbn}/authors/{authorId}

DELETE /api/v1/books/{isbn}/authors/{authorId}

Publisher APIs

HTTP

GET /api/v1/publishers

GET /api/v1/publishers/{publisherId}

POST /api/v1/publishers

PUT /api/v1/publishers/{publisherId}

DELETE /api/v1/publishers/{publisherId}

Inventory APIs

HTTP

GET /api/v1/inventory

GET /api/v1/inventory/{inventoryId}

GET /api/v1/inventory/books/{isbn}

POST /api/v1/inventory

PUT /api/v1/inventory/{inventoryId}/purchase

Book Condition APIs (Read-Only)

HTTP

GET /api/v1/book-conditions

GET /api/v1/book-conditions/{rank}

User APIs

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HTTP

POST /api/v1/users

GET /api/v1/users/{userId}

PUT /api/v1/users/{userId}

Role APIs (Admin)

HTTP

GET /api/v1/roles

PUT /api/v1/users/{userId}/roles/{roleId}

Purchase / Order APIs

HTTP

POST /api/v1/orders/checkout/{userId}

GET /api/v1/orders/user/{userId}

Shopping Cart APIs

HTTP

GET /api/v1/cart/{userId}

POST /api/v1/cart/{userId}/items/{isbn}

DELETE /api/v1/cart/{userId}/items/{isbn}

Review APIs

HTTP

GET /api/v1/books/{isbn}/reviews

POST /api/v1/books/{isbn}/reviews

Reference APIs (Read-Only)

HTTP

GET /api/v1/states

GET /api/v1/roles

High-Impact Report Endpoints

✓ 1. Top Selling Books

HTTP

GET /api/v1/reports/books/top-selling?limit=5

-Why this is important

- Core business KPI (revenue driver)
- Requires joins + aggregation
- Tests real-world thinking (ranking, sorting)

-Data Logic

- Use inventory + purchaselog
- Count how many inventory items were purchased per book

-Sample Response

JSON

```
[
{
  "isbn": "1-111-11111-4",
  "title": "Women are From Venus...",
  "soldCount": 25
}
```

-What you evaluate

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- GROUP BY + COUNT
- Sorting (ORDER BY soldCount DESC)
- DTO mapping

✓ 2. Sales by Category (Business Insight)

HTTP

GET /api/v1/reports/sales/by-category

-Why this is important

- Shows analytical thinking
- Connects multiple tables (book → category → inventory)
- Used in dashboards & reporting tools

-Data Logic

- Join:
 - book.Category
 - category
 - inventory (Purchased=1)

-Sample Response

JSON

```
[  
  { "category": "Programming", "booksSold": 40 },  
  { "category": "Romance", "booksSold": 30 }  
]
```

-What you evaluate

- Multi-table joins
- Business grouping
- Clean response modeling

✓ 3. Book Rating Summary (Customer Insight)

HTTP

GET /api/v1/reports/books/{isbn}/rating-summary

-Why this is important

- Evaluates customer feedback understanding
- Requires aggregation & null handling
- Real-world (used in Amazon/Flipkart style apps)

-Data Logic

- Use bookreview
- Calculate:
 - AVG(rating)
 - COUNT(*)

-Sample Response

JSON

```
{  
  "isbn": "9-455-61223-3",  
  "averageRating": 8.5,  
  "totalReviews": 10  
}
```

-What you evaluate

- Aggregate functions
- Handling no-review cases
- REST design clarity

Book Inventory Application

Why:

Endpoint	Covers
Top Selling Books	Sales analytics + ranking
Sales by Category	Business reporting + joins
Rating Summary	Customer insights + aggregation